

⇒ Types of semiconductor materials:-

① Intrinsic

② Extrinsic

Intrinsic → Pure semiconductor material

Extrinsic → Impure semiconductor material

1) N-Type (Pentavalent)

2) P-Type (Trivalent)

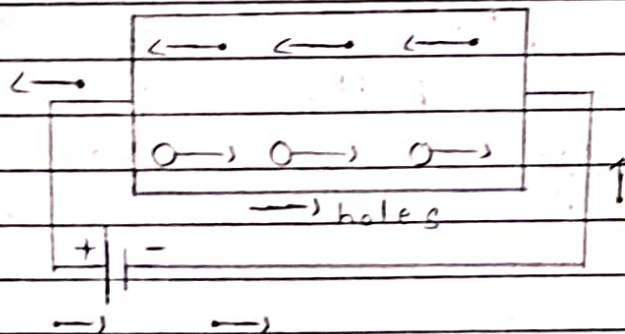
Intrinsic Semiconductor:-

→ A semiconductor in a extremely pure form is known as intrinsic semiconductor material.

→ In an intrinsic semiconductor, even at room electron lone pairs are created.

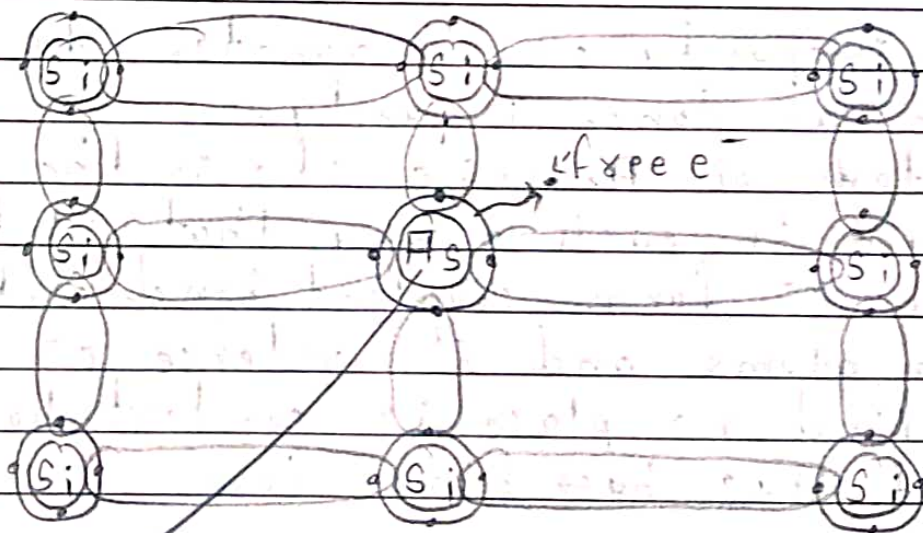
→ Under the influence of electric field conduction to the semiconductor is by both free e^- & holes.

→ Therefore the total current inside semiconductor is the sum of current due to free e^- & holes.
is zero



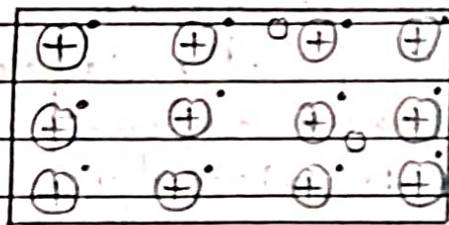
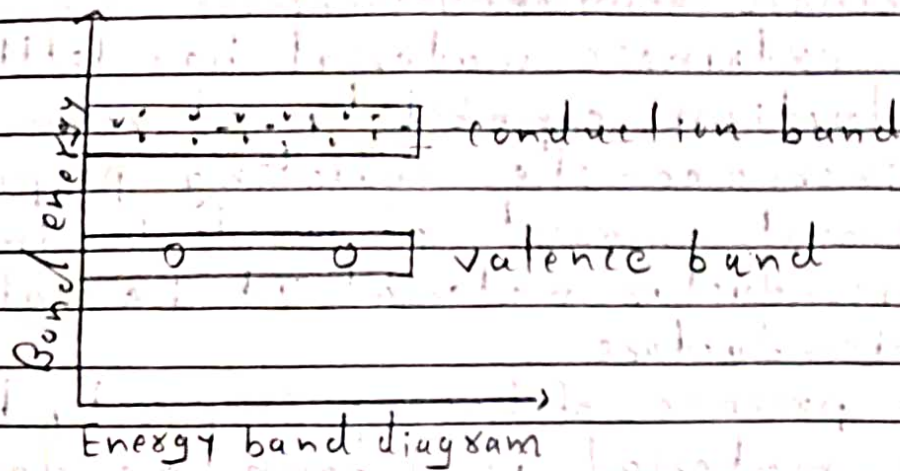
Extrinsic semiconductors:-

- The extrinsic material has little current conduction capability at room temperature to increase its conducting property we had a small amount of suitable impurities the semiconductor material. It is called extrinsic semiconductors.
- The process of having impurity to a semiconductor material is known as doping.
- Generally for 10^8 atoms of semiconductors one impurity atom is added.
- If pentavalent impurity is added to a semiconductor. A large amount of free e^- are produced in semiconductor & it is called N-type semiconductor material.
- If we added trivalent impurity to a semiconductor then of large no. of holes is produced and it is called p-type semiconductor material.
- N-type semiconductors materials:-



Impurity Atom

Structure of N-type semiconductor



N-Type

It consist

- e^-
- holes
- ions

- When small amount of pentavalent impurity is added to a pure semiconductor it is known as N-Type semiconductor materials.
- N-type impurities are As-at.no. - 33

Antimony - at. no. - 51

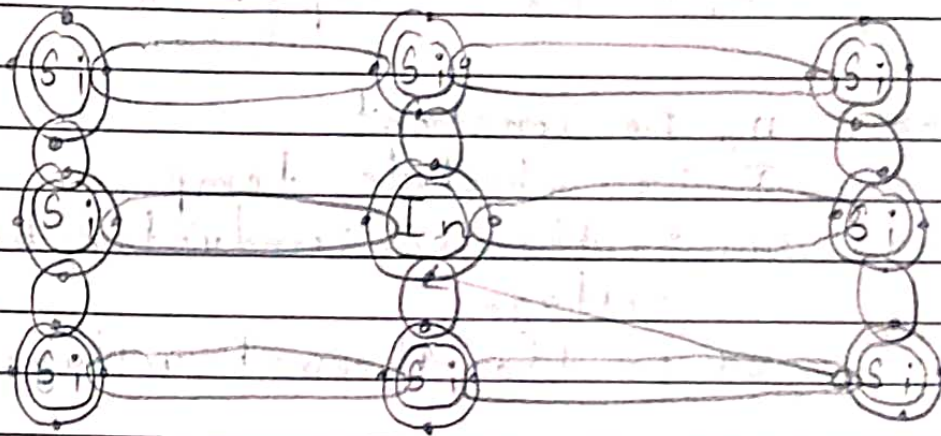
Such impurity which donates its e^- known as donor impurity.

- The atom of Arsenic is in the silicon structure in such a way that its four valence e^- form covalent bond with a silicon atom and 5th valence e^- of As atom find no place in covalent bond and it thus becomes free.

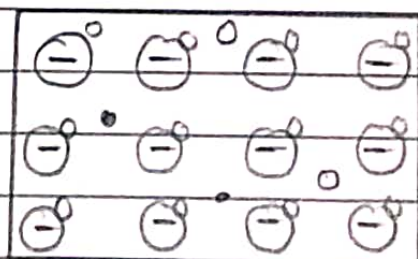
→ The energy band diagram shown in figure has a no. of conduction band e^- that is free e^- and valence band has few holes generated by thermal energy of room temp.

→ An N-type material has a no. of e^- free hole and +ve ion generated by impurity atoms.

⇒ P-type semiconductors:-



- ① Hole created by doping
- ② electron-hole pair due to room temperature.
- ③ -ve ions.



P-type semiconductor.

Concentration of Intrinsic semiconductor:

→ In an intrinsic semiconductor the hole concentration must be equal to electron concentration. In this case we write

$$n = p = n_i$$

where,

n_i → Intrinsic conc.

and n_i is denoted as

$$n_i = n_0 T^{3/2} \exp\left(\frac{-V_{G1}}{2V_T}\right)$$

where n_0 is constant

T is absolute temp.

V_{G1} is the semiconductor bandgap voltage.

V_T is the thermal voltage.

$$\text{Also, } V_T = \frac{kT}{q}$$

k = Boltzmann constant

T = thermal voltage

q = charge

⇒ Conductivity:-

→ When a electric field E is apply to an intrinsic semiconductor then free electron drift with the velocity.

$$\vec{V}_e = -\mu_e \vec{E}$$

$\mu_e \Rightarrow$ electron mobility

→ The holes drift with a velocity

$$\vec{V}_h = +\mu_h \vec{E}$$

→ Although the free electron and holes drift in opposite direction the current density is add because charge polities is opposite.

→ The electron charge density is

$$\rho_e = -n_i q$$

and the holes charge density is

$$\rho_h = +n_i q$$

→ When n_i is intrinsic concentration.

→ The total conduction current density can be written as

$$\begin{aligned}\vec{J} &= \rho_e \vec{V}_e + \rho_h \vec{V}_h \\ &= n_i q \mu_e \vec{E} + n_i q \mu_h \vec{E} \\ &= n_i (\mu_e + \mu_h) q \vec{E}\end{aligned}$$

$$\boxed{\vec{J} = \sigma \vec{E}}$$

(Q) A rod intrinsic silicon is 1cm long and has a diameter of 1mm. At room temperature intrinsic concentration in the silicon is $1.5 \times 10^{16}/m^3$. The electron and hole mobility are $0.13 m^2 V^{-1} s^{-1}$ and $0.05 m^2 V^{-1} s^{-1}$. Calculate the conductivity of silicon and the resistance of rod.

→

Data:-

$$\sigma = n_i (\mu_e + \mu_h) q$$

$$= 1.5 \times 10^{16} (0.13 + 0.05) \times 1.6 \times 10^{-19}$$

$$\boxed{\sigma = 0.432 \times 10^{-3} S/m}$$

$n_i = 1.5 \times 10^{16}/m^3$
 $\mu_e = 0.13 m^2 V^{-1} s^{-1}$
 $\mu_h = 0.05 m^2 V^{-1} s^{-1}$
 $d = 1 mm$
 $L = 1 cm$

$$R = \frac{\rho L}{A}$$

$$\rho = \frac{1}{\sigma}$$

$$= \frac{1}{0.432 \times 10^{-3}} \times 10^{-2}$$

$$\rho = \frac{1}{0.432 \times 10^{-3}}$$

$$3.14 \times 0.00000025$$

$$R = 29,488,086.81$$

$$A = \pi \times r^2$$

$$= 3.14 \times \left(\frac{0.001}{2}\right)^2$$

$$\boxed{R = 29.4 \text{ mega } \Omega}$$

$$= 3.14 \times 0.00000025$$

⇒ Conductivity in extensive material:-

① Mass action law:-

- In an intrinsic semiconductor material, the electron concentration and the hole concentration are both equal to intrinsic concentration.
- That is $n = p = n_i$
- When we add N type of impurities to the semiconductor material number of electron increase and holes decrease
- Similarly adding a p-type impurity increases p and decreases n .
- So the product N times p is a constant independent of the doping type and level.
- The product is given by

$$n \cdot p = n_i^2 \quad \text{--- (F)}$$

and this relation is called mass action law.

② Electrical Neutrality:-

- An intrinsic semiconductor is electrically neutral that is no net charge store.
- The addition of N-Type or P-Type impurity does not change.
- Let N_D with a no. of donor atom per m^3 and N_A the no. of acceptor atom per m^3 .
- The total no. of negative charge per m^3 is equal to the no. of free electron and no. of acceptor atom N_A per m^3 .

$$n + N_A$$

→ Similarly the no. of positive charge per m^3 is equal to the no. of holes and the no. of donor atom N_D per m^3 .

$$p + N_D$$

→ Now because the semiconductor is electrically neutral the no. of positive charge is equal to no. of negative charge.

$$n + N_A = p + N_D \quad \text{--- (1)}$$

→ In an N-Type semiconductor $N_A = 0$

$$n = p + N_D$$

and

$$p \ll n$$

$$n \approx N_D \quad \text{--- (2)}$$

→ The approximation in eqⁿ - (1)

$$p = \frac{n_i^2}{n}$$

$$\boxed{p \approx \frac{n_i^2}{N_D}}$$

→ Similarly in P-Type material we can write

$$P = n + NA$$

$$n \approx \frac{n_i^2}{NA}$$

Conductivity:-

→ The conductor current density in an extrinsic semiconductor:

$$\vec{J} = n_i (\mu_e + \mu_h) q \vec{E}$$

→ In a doped semiconductor material it is given by

$$\vec{J} = (n_e \mu_e + n_h \mu_h) q \vec{E}$$

→ Where n_e is the electron concentration and n_h is the hole concentration. so the eqn defines the conductivity.

$$\sigma = (n_e \mu_e + n_h \mu_h) q$$

The hall effect:-

→ The Lorentz force on the electrons is given by

$$F_M = -BeV_d$$

where

e → magnitude of charge

V_d → drift velocity

An electric field formed, is given by

$$F_e = -eE_h$$

E_h → electric field due to the hall effect

F_e → force due to the electric field

→ In neutral situation Lorentz force and electric force is balanced.

Therefore

$$F_M = F_e$$

$$-BeV_d = -eE_h$$

$$\Rightarrow E_h = BV_d \quad \text{--- (1)}$$

→ Let b be the length of the cross section then the electric field generated due to hall effect is given by

$$E_h = V_h / b$$

$$\text{or } V_h = E_h \cdot b$$

where V_h is the hall voltage

$$V_h = B V_d b \quad - (2)$$

→ Let J be the current density then it is given by

$$J = -n_e e \times V_d$$

$$V_d = \frac{-J}{n_e e}$$

substituting the above eqⁿ in eqⁿ (2)

$$V_h = \frac{-B b J}{n_e e}$$

→ Let d be the width of the cross section of the conductor then the current density.

$$J = \frac{I}{A} = \frac{I}{d \cdot b}$$

$$\therefore V_h = \frac{-B b I}{n_e e d b}$$

$$\boxed{V_h = \frac{-B I}{n_e e d}}$$

→ These is the expression for the hall voltage that develops across the conductor due to hall effect

$$V_h = \frac{-BI}{ne d}$$

also we have $V_h = E_h b$

$$E_h = \frac{V_h}{b}$$
$$= \frac{-BI}{ne d b}$$

$$E_h = \frac{-BJ}{ne}$$

→ The above expression is known as the hall field.

→ From the above equation we get

$$\frac{E_h}{BJ} = \frac{1}{ne}$$

→ The ratio between the hall field to the product of magnetic field B and current density J is known as hall coefficient.

$$R_h = \frac{1}{ne}$$

→ When the no. of electrons exceeds the no. of holes the hall coefficient

- formula become negative.
- ⇒ Hall derivation in semiconductor:-
- Because electrons and holes contribute to diffusing conc. and mobilities in semiconductor it is difficult to explain the hall coefficient given above as a result hall coefficient is as follows for a simple explanation of a moderate magnetic field.

$$R_h = \frac{p \mu_h^2 - n \mu_e^2}{e (p \mu_h + n \mu_e)}$$

where p is the hole conc.

n is the electron conc.

μ_h is the mobility of hole

μ_e is the mobility of electron

e is the electron charge.

⇒ Application of Hall effect:-

- It determines by the whether a particular material is a semiconductor or a insulator.
- It is used magnetometer to measure the magnetic field.
- It is used in position sensing because they are resistant to water, mud, dust and grime.
- It is used a hall effect sensor in integrated circuit.
- It detects wheels speed and assists the

antilock breaking system accordingly.

⇒ Nanomaterials :-

1) NanoScience :-

- is a phenomena of manipulation of materials at the atomic molecular and micromolecular scales. where properties differ significantly from those at larger scale.

2) Nanomaterials :-

- are defined as particles in the form of crystal, rods or spheres having size between $1 \times 10^{-9} \text{ m}$ and $100 \times 10^{-9} \text{ m}$ at least in one dimension.
- A nanometre is one billionth of metre $= 10^{-9} \text{ m}$.

⇒ Properties of nano-materials :-

- Physical properties that make nanomaterials different from bulk materials.

1) Due to smallness of nanomaterial gravitational forces become negligible.

2) For objects of very small mass such as electron wave particle duality has more pronounced effect. Quantum mechanics is used to describe motion and energy instead of classical mechanics.

3) The consequence of this is the tunneling. It is the penetration of an electron into an energy ~~region~~ region that is classically forbidden.

4) Quantum confinement:-

→ In a nanomaterial such as a metal electrons are confined in space rather than free to move in the bulk of the material.

5) Quantisation of energy:-

→ Electrons in a nanomaterial can exist at discrete energy levels.

6) At nanoscale, the random motions are of same scale as the size of the material. This has an influence on how particle behaves.

7) Increase surface to volume ratio.

⇒ Properties synthesis of nanomaterials:-

→ There are two general approaches to the synthesis of nanomaterials.

(1) Top-down method:-

→ This approach uses larger (macroscopic) initial structure which can be extremely controlled in the processing of nanostructures.

1) (2) Top-down lithography :-

→ The most used top down approach is photolithography. It has been used to manufacture comp. chips and produce structures smaller than 100 nm. Typically, an oxidized silicon (Si) wafer is coated with a 1 μ m thick photo resist layer. After exposure to ultraviolet (UV) light, the photoresist undergoes a photo chemical reaction, which breaks down the polymer by capturing the polymer chains. Subsequently, when the wafer is rinsed in a developing solution, the exposed areas are removed leading to nanosize material. The another methods are - electron beam and x-ray lithography.

→ Ballmilling process:- It used for grinding, mixing and mechanical alloying of materials. It involves a rotating drum filled with grinding balls that with the material to be milled resulting in reduced particle size and mixing. It's widely used in materials science and technology for various applications.

2) Bottom-up approach:- This approach include the miniaturization of materials components with further self assembly process leading to the formation of nanostructures.

- E.g. 1) Sol gel processing 4) laser pyrolysis
2) chemical vapour deposition
3) Plasma or flame spraying synthesis
5) Inert gas condensation.

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1) Sol gel technique :- creating a solⁿ of metals alkoxides or salts, forming nanoparticles suspended in a liquid solⁿ.

(b) Gel formation : chemical reaction cause nanoparticles to link and form a three-dimensional network.

(c) Drying :- Removing the solvent to create a solid materials with a porous structure.

(d) Optional Calcination :- Heating the material to further modify its structure.

2) chemical vapour deposition process :- Vapour is formed in a reaction chamber by pyrolysis reduction, oxidation or nitridation, and then deposited on the surface.

3) The inert gas condensation : This process is are most known and simple technique for production of nanoparticles. An inorganic material is vapourized inside a vacuum chamber into which an inert gas is periodically admitted.

4) The production route for 1-D rod-like nanoparticles materials by liquid phase methods is similar to that for the production of nanoparticles.

classification of nanomaterials:-

→ Range is $< 100 \text{ nm}$

① 0-D:-

→ All dimensions at nanoscale.

② 1-D:-

→ Two-dimensions at the nanoscale and one-D at macroscale.

③ 2-D:-

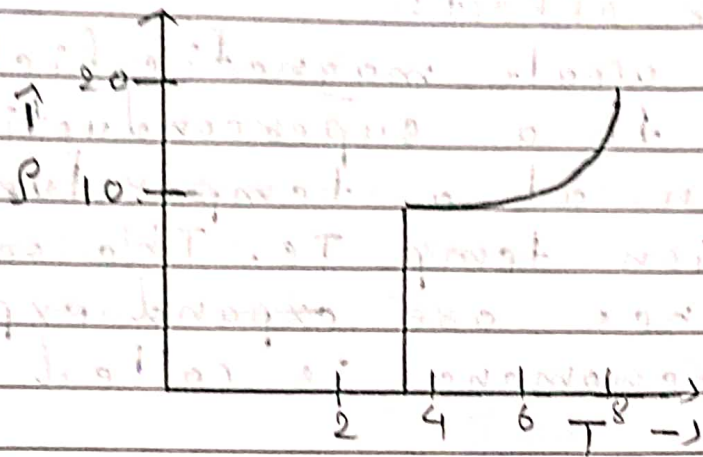
→ One-D at the nano-scale and two-D at the macroscale.

④ 3-D:-

→ No-D at nanoscale and All-D at macroscale.

⇒ Super Conductivity :-

- Certain metals and alloy exhibits almost zero resistivity when they are cooled to significantly low temperatures. This effect is called super conductivity.



- Critical or Transition temperature :-
- The temp. at which the transition from normal state to super conductivity state take place on cooling in the absence magnetic field is called critical temp. or transition temp.
- Properties of Super Conductivity :-
- It is a low temp. phenomena.
- Transition temp is different for different substances.
- Material having high normal resistivity, exhibits super conductivity.
- Material for which $Z_p = 10^6$ show super conductivity.
- Super conductivity is very sharp for chemically pure or structurally perfect specimen.

- Ferromagnetic and Antiferromagnetic materials are not superconductors.
- Superconductors elements lies in the inner column of periodic table.

⇒ Meissner effect:-

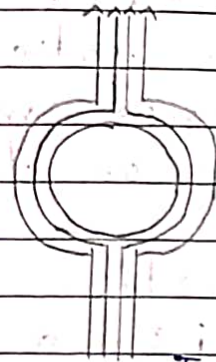
- Where a weak magnetic field is applied to a superconducting specimen at a temperature below transition temp T_c . The magnetic flux lines are expelled. This phenomena is called Meissner effect.

- Under normal state the magnetic induction inside the specimen is

$$B = \mu_0(H + I)$$



$T > T_c$



$T < T_c$

where $H \rightarrow$ Applied magnetic field

$I \rightarrow$ magnetization produced inside the specimen.

when the specimen in superconductors state $B = 0$ (Meissner effect)

$$C = \mu_0 (H + I)$$

$$\text{or } H = -I$$

$$\text{or } \chi = \frac{I}{H} = -1$$

- Does the material act as a perfectly diamagnetic. (for $\chi = -1$)
- Let us consider a superconducting material in normal condition from ohm's law the electric field is

$$E = J \rho \rightarrow \text{Resistivity}$$
 where J is current density.
- on cooling the material to its transition temperature $\rho \rightarrow 0$
- If J is have finite E must be zero.
- From Maxwell eqⁿ

$$\nabla \times E = - \frac{dB}{dt}$$

under superconducting condition since $E = 0$

$$\frac{dB}{dt} = 0 \text{ or } B \text{ is constant.}$$

- This means that the magnetic flux passing through the specimen should not change on ~~pulling~~ cooling to the transition temp.
- The Meissner effect contradicts the result.

⇒ Types of superconductors:-

→ Based on the diamagnetic response superconductors are classified into two categories:

① Type - I

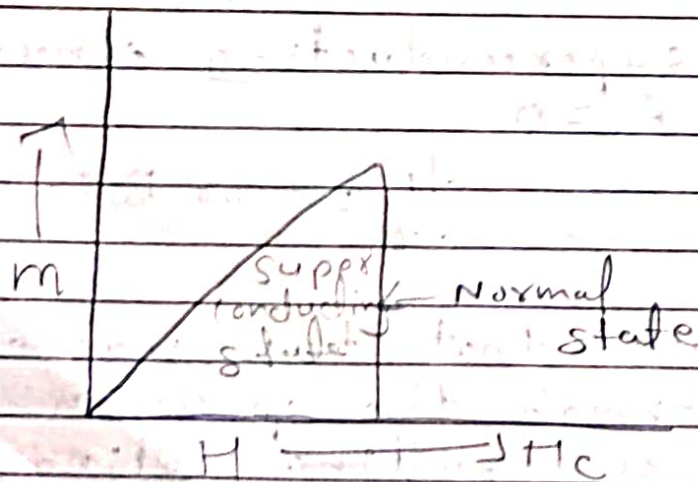
② Type - II

① Type - I :-

→ Superconductors which follow a complete Meissner effect is called type I superconductors (also known as soft superconductors).

→ When the magnetic field strength is gradually increased from its initial value $H < H_c$, at H_c the diamagnetism is abruptly disappears and the transition from superconducting state to normal state is sharp as shown in figure. These superconductors are known as soft or superconductors.

→ Examples:- Al, Zn, Hg and Sn



② Type II:-

- Superconductors which does not follow the complete Meissner effect is called type I superconductors (also is known as hard superconductors).
- In type II superconductors, the specimen is in pure superconducting state up to the field H_{c1} (lower critical field) when the field is increased beyond H_{c2} (upper critical field) the magnetic flux lines start penetrating. The specimen is in mixed state between H_{c1} and H_{c2} . Above H_{c2} , the specimen is in normal state.
- This means that the Meissner effect is incomplete in the region between H_{c1} and H_{c2} .
- This region is known as vortex region. These superconductors are known as hard superconductors.
- Examples:- Zr , Nb

